# MODULE: MACROMOLECULAR ANALYSIS

## Proteins – Structure and Function:

Short chains of amino acids are called peptides. The chains which have fewer than 40 or 50 amino acids are called polypeptides, while still larger chains are called proteins. They consist of carbon, hydrogen, oxygen, nitrogen elements, and some others may contain sulphur, phosphorus, iron and other elements.
Proteins may have many roles in the body and are largely responsible for the structure of body tissues.

## Hierarchy in protein structure:

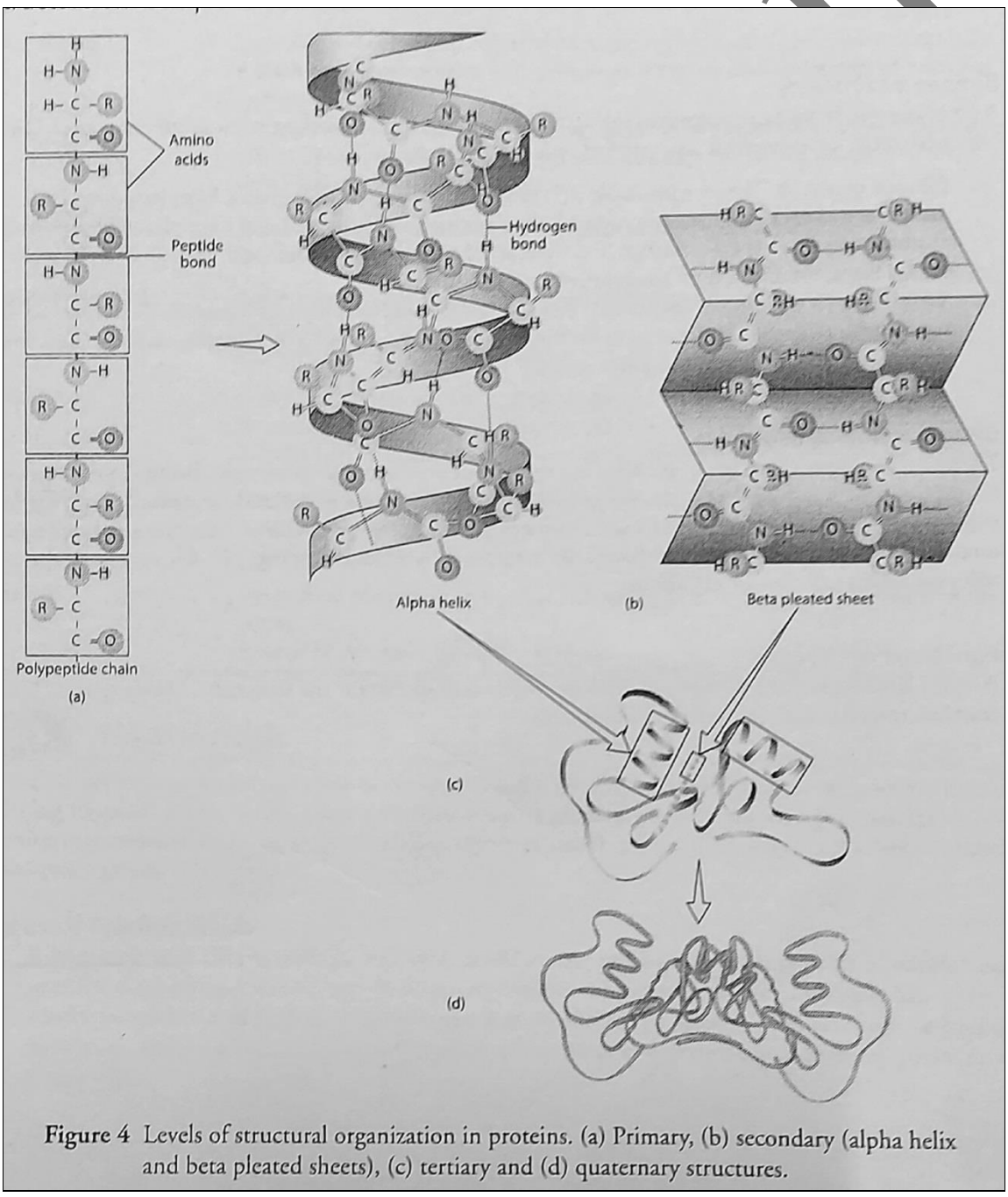



**Figure 4** Levels of structural organization in proteins. (a) Primary, (b) secondary (alpha helix and beta pleated sheets), (c) tertiary and (d) quaternary structures.

1. **Primary structure:** It is determined by the linear sequence of amino acid molecules in a polypeptide chain, reading from N-terminus to C-terminus.
2. **Secondary structure:** A functional protein may contain one or more polypeptide chains. Secondary structure is determined by H-bonds between the carboxyl oxygen and hydrogen atom of component amino group. These bonds can occur either between different polypeptide chains of the protein or within the molecules of the same polypeptide chain. Secondary structure is determined by:
    a. **α-helix:** Polypeptide chain coiled spirally in a right handed manner, stabilised by H-bonds between oxygen of carbonyl group of one amino acid and hydrogen of amino group of fourth amino acid residue next to former.
    b. **β-sheet:** in this case, two or more polypeptide chains are joined together by H-bonds between oxygen of carbonyl group of one amino acid and hydrogen of amino group of another amino acid of adjacent chain. The adjacent polypeptide chain may run in the same direction (parallel) or in opposite direction (antiparallel).
3. **Tertiary structure:** it involves twisting and folding of the secondary structure of a polypeptide chain upon themselves to produce stable 3D-structure. The 3D-structure is maintained by four types of bonds, they are:
    a. **Ionic bonds:** Developed between two polar groups.
    b. **Hydrogen bonds:** Formed between two groups which are linked through a shared hydrogen atom.
    c. **Disulphide bonds:** Formed between two cysteine residues.
    d. **Hydrophobic bonds:** Developed between two non-polar groups.
4. **Quaternary structure:** It is found in oligomeric proteins. Such proteins are formed of more than one polypeptide chains, each chain having its own primary, secondary and tertiary structure. Each polypeptide chain functions as subunit of the protein. The different subunits are packed together to give the final 3D conformation of the oligomeric protein. The quaternary structure is stabilised by the non-covalent forces as in case of tertiary structure.

## Protein classification on the basis of function:

1. **Enzymes:** Such proteins act as biocatalysts, each capable of catalysing a different chemical reaction. They are responsible for metabolic regulation and thus control physiological activities of the cell. E.g. – Oxidoreductase, Transferase, Hydrolase, Lyase, Isomerase, Ligase.
2. **Transporters:** Such proteins act as physical carriers of other functionally important smaller molecules. E.g. – Permease transport specific molecules or ions across the cell membrane. Haemoglobin transport respiratory gasses. P-protein helps in transport of organic compounds through phloem. Serum albumin transports fatty acids and lipids in blood.
3. **Receptors:** Such proteins are present on cell membrane. They bind specific hormones or neurotransmitters and make the cell receptive of their effect. E.g. – Hormone receptors like FSH-receptors, epinephrine receptors, oxytocin receptors, etc.
4. **Structural elements:** Such proteins serve as building material of cells and tissues. These are essential for protection, growth and repair. E.g. – Matrix of connective tissue contains collagen and elastin protein. Matrix of cartilage contains chondrin protein. Matrix of bone contains ossein protein. Protective covering of body like skin, scales, nails, hair, hoof, etc. contains keratin protein.

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## Protein classification on the basis of shape:

1. **Fibrous proteins:** They are insoluble in water and their polypeptide chains form long strands that are parallel to each other. Fibrous proteins have many structural functions. For example, collagen strengthens bones, ligaments and tendons. Keratin forms structure of hair, nails and waterproofs the skin.
2. **Globular proteins:** They are more or less soluble in water and their polypeptide chains are spherical (globular) in shape. Globular proteins have metabolic functions. For example, enzymes function as catalysts and haemoglobin transports oxygen.

## Protein denaturation:

The three-dimensional shape of proteins is dependent on the temperature and chemical composition of body fluids which in turn are regulated by homeostatic mechanisms. If the environment changes, a protein may lose its characteristics shape. This process is called denaturation. The factors that affect the denaturation of proteins include: heat, pH, detergents, oxidising agents, reducing agents and denaturing agents.

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